

Fundamental parameters of Ap-stars

78 Vir (HD 118022) and AX CVn (HD 110066)

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- Objects of study: HD 118022 (78 Vir) and HD 110066 (AX CVn);
- The goal;
- Derived chemical abundance;
- Selection of the atmosphere parameters T_{eff} and $lg g$;
- Stratification analysis of Fe and Cr for 2 Ap-stars;
- Evolutionary status;
- Conclusion.

HD 118022 DATA

- Variable Star of α^2 CVn type, A1p CrSrEu;
- $P_{rot} = 3^d.7220$;
- $\pi = 17.1618 \pm 0.2286 mas$ - GAIA DR2
- $B_s = 3000 Gs$;
- NARVAL spectropolarimeter (phases 0.095, 0.162, 0.211, 0.371, 0.490, 0.623 and 0.721). $R = 65000$, spectral range: 3700–10000 Å;
- UVES - phase 0.947, $R = 80000$, spectral range: 3100–10000 Å;

OBJECTS OF STUDY

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HD 110066 DATA

- Variable Star of α^2 CVn type, A0p SrCrEu;
- $\pi = 7.1310 \pm 0.0561 \text{ mas}$ - GAIA DR2
- $B_s = 4100 \text{ Gs}$;
- ESPaDonS spectrometer. $R = 81000$, spectral range: 3700–10000 Å;

OBJECTS OF STUDY

THE GOAL

- Chemical abundance analysis.
- Stratification of Fe and Cr in their atmospheres.
- Determination of the fundamental parameters.
- Their position on the Hertzsprung-Russell Diagram.

DETERMINATION OF T_{eff} AND $lg g$ BY SPECTROSCOPY

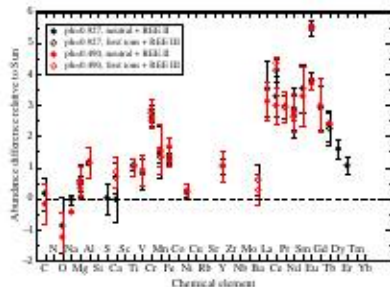
Determination of fundamental parameters is an iterative process:

- 1 Calculation of model atmosphere with the initial parameters T_{eff} and $lg g$.
- 2 Calculation of synthetic spectrum with magnetic field. Atomic data were derived from VALD3.
- 3 Abundance determination.
- 4 Stratification calculation for Fe and Cr elements.
- 5 Refinement of atmosphere parameters through comparison of the calculated flux with the observed spectral energy distribution (SED).
- 6 If in comparison with the observed fluxes there remain a discrepancy, then these steps are repeated, specifying the model atmosphere until the discrepancy become minimal.

For HD 118022 were made 3 iterations, for HD 110066 - 3 iterations.

CHEMICAL ABUNDANCES

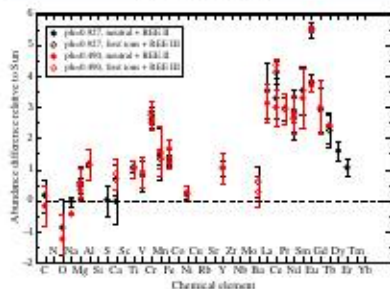
HD 118022



Chemical abundances in 2 phases of rotations relative to solar values.

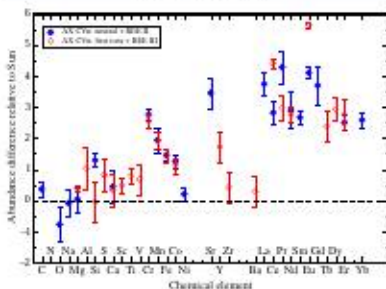
CHEMICAL ABUNDANCES

HD 118022



Chemical abundances in 2 phases of rotations relative to solar values.

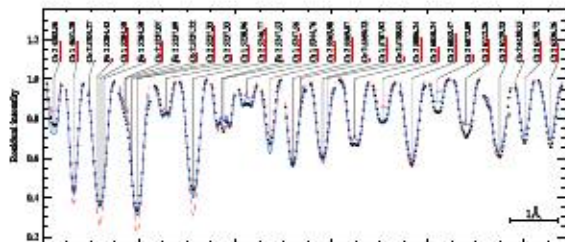
HD 110066



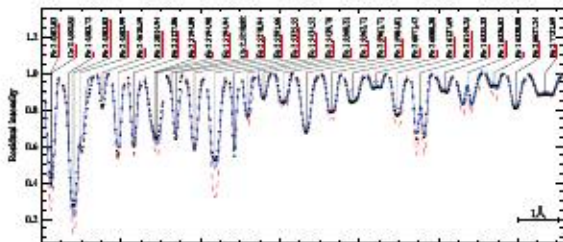
Chemical abundances relative to solar values.

STRATIFICATION LINES FOR HD 110066

Cr lines – 19



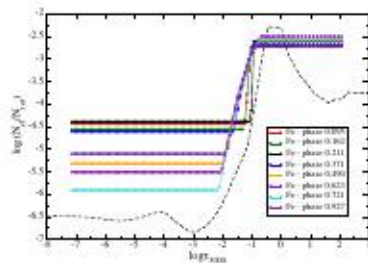
Wavelength



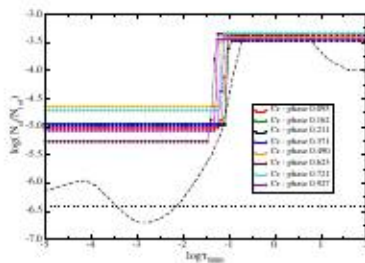
Wavelength

Fe lines – 24

STRATIFICATION



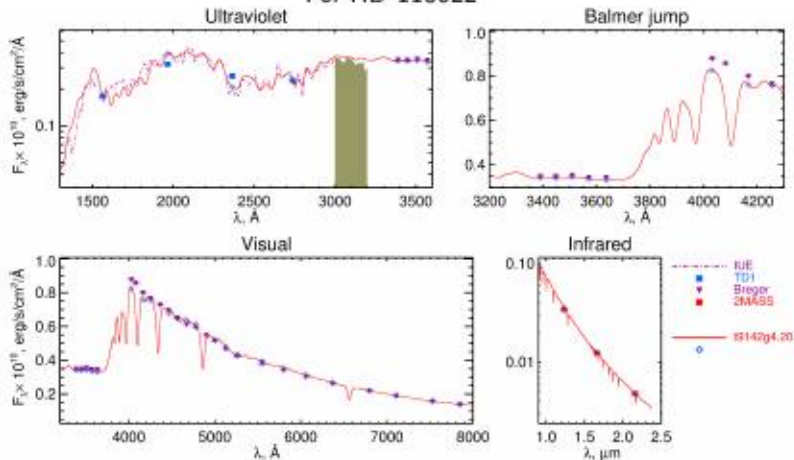
HD 118022 Fe stratification



HD 118022 Cr stratification

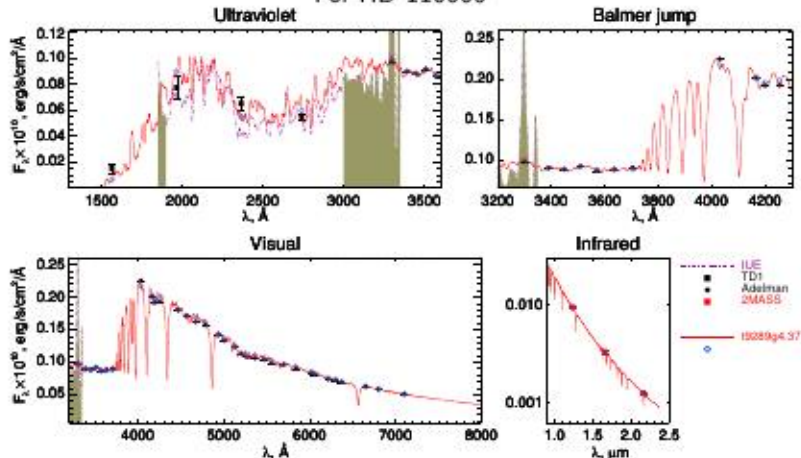
CHOOSING T_{eff} AND $lg g$

For HD 118022

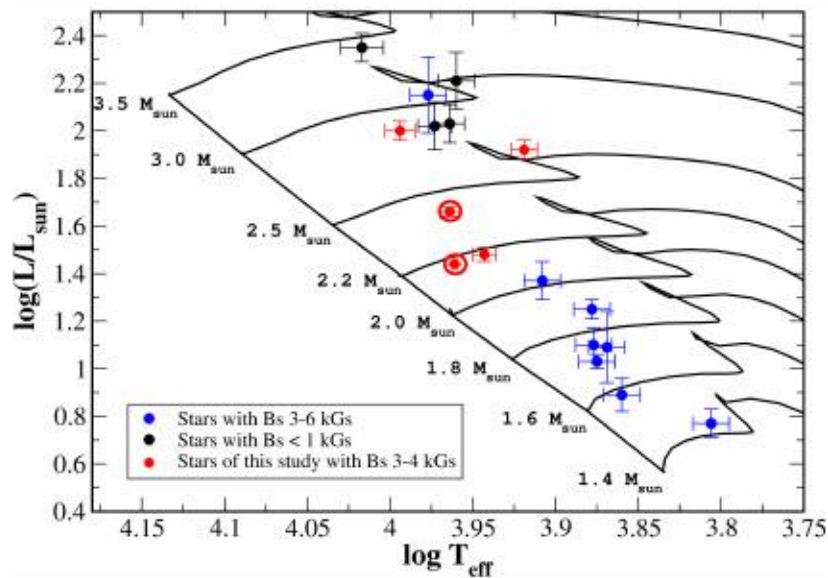


CHOOSING T_{eff} AND $\lg g$

For HD 110066



EVOLUTION STATUS



CONCLUSION

- 1 HD 118022 and HD 110066 show nearly similar element abundance pattern: CNO deficiency, practically solar Na and Mg abundance, 1-2 dex overabundance of the iron peak elements, and a large excess of the rare-earth elements (REEs).
- 2 The vertical distribution (stratification) of Fe and Cr elements for all spectra were derived. And for both stars Cr and Fe have sharp abundance gradients with abundance jumps at optical depth nearly $\log \tau_{5000} \sim -1.75$;
- 3 The atmospheric analysis of the these two Ap stars results in the determination of the effective temperatures T_{eff} , surface gravities $\lg g$, radii $R/R_{\odot}$ and luminosities $L/L_{\odot}$, as well as element abundances by self-consistent way. Error of the measurement in the last digits is given in parentheses.

Stars	T_{eff} , K	$\lg g$	B_s , kG	$R/R_{\odot}$	$L/L_{\odot}$
HD 118022	9142(30)	4.2(2)	3.0	2.10(2)	1.44(5)
HD 110066	9200(100)	4.1(1)	4.1	2.68(2)	1.66(2)

CONCLUSION

Thanks for your attention!